06. Business, Unit Economics and Scaling

AdmitAssist | MBA ZG583 Management of AI Products | GOLLAPUDI SIVA NAGA JNANA TEJA | 2025MB26205 | GitHub: jnanateja009

1. Unit economics in one minute

A school pays **Rs 1,499 a month in admission season (December to May) and Rs 299 a month in the quiet months**: Rs 10,788 a year. Reading and drafting one inquiry costs me about **Rs 0.36 in AI calls**. In a busy month with 200 inquiries, my cost to serve that school is about **Rs 372** (AI Rs 72, support Rs 250, payment fee Rs 30, storage Rs 20), so I keep **Rs 1,127, a 75% contribution margin**. Winning and setting up a school costs about Rs 6,000 of my time, paid back in about 5 busy months. Three schools cover the shared server; 78 schools would pay me Rs 50,000 a month. For the school, **one extra admission a year pays for the tool**. The risk is not the AI bill. It is support time: at 3 hours per school per month I lose money.

2. Who pays, and for what

Stakeholder	Pays?	What they are really buying
School owner	Yes, monthly	Not losing serious families to a slow or wrong reply; a record of every inquiry; not hiring a second front-office person
Front-office coordinator	No	Fewer hours of reading and typing in peak weeks; protection from giving wrong answers
Principal	No	Sensitive cases reach a person
Parent	No, never	A correct answer the same working day (HIGH inquiries within 2 working hours)

The faculty's point that people buy a safety net more than a product fits here: the owner is buying "no serious parent slips through", not software.

3. Price: worked backwards from value

I used target costing (start from what the school will pay, subtract my margin, and see what I may spend), not cost-plus.

Step	Figure	Basis
What one admission is worth to a budget private school	about Rs 12,000 a year in fees	About 70% of private-school parents pay under Rs 1,000 a month (SOURCE: Central Square Foundation, State of the Sector Report on Private Schools, 2020)
What an owner will pay for "not losing one admission"	up to about one admission's fees a year	DECISION: the price must pay for itself with one extra admission
Yearly price	Rs 10,788 (0.90 of one budget-school admission)	6 x Rs 1,499 + 6 x Rs 299
Target margin	70% or more in season	DECISION
So the most I may spend to serve one school in a busy month	about Rs 450	30% of Rs 1,499
What I actually spend (base case)	Rs 372	Section 5

Staff time saved does **not** justify the price on its own: about 8.0 hours a month worth Rs 769 at a Rs 20,000 salary (SOURCE: AmbitionBox average of Rs 2.4 lakh a year for receptionist and front-office executive). I say this openly in the pitch. The case is about admissions.

3.1 Pricing model and why

Option	Verdict
Per token or per AI call (input-based)	No. The school pays for my prompt mistakes and retries, which the faculty rightly called unfair.
Flat monthly all year	No. Paying full price in July for an empty inbox is the "why pay during Diwali" problem. Schools would cancel in June.
Seasonal subscription with a pause plan (chosen)	Rs 1,499 per admission-season month with 300 inquiries included; Rs 299 per quiet month with 50 included; Rs 2 per extra inquiry. The school can switch plan for any month. Quiet months bring in little on purpose (Rs 212 contribution a month), to keep the school through the year (the "downsell to retain" idea from class).
Outcome-based (per admission)	Not in V1. I cannot yet prove which admissions AdmitAssist caused. V3 trial once the funnel report (F12) can link an inquiry to an admission: Rs 500 per admitted child whose first inquiry was handled in AdmitAssist, capped at Rs 15,000 a year.
Pilot school	Free for the 2027-28 season (21 Dec 2026 to 31 May 2027) in exchange for anonymised data, weekly feedback and a reference. Not counted as revenue.

4. Competition

Option a school has today	What it does	Price (public)	Why a small school does not use it for this job
Paper register + WhatsApp Business app	Free messaging, labels	Free	No reading, no priority, no checks; register skipped in peak weeks
Staff paste into a free AI chat	Drafts replies	Free tier	Invents fees and dates (no school facts), no priority or log, parents' numbers go into personal accounts
WhatsApp platforms: AiSensy, Interakt, Wati	Shared inbox, broadcasts, chatbots that reply on their own, AI agents as add-ons	Rs 1,350 (AiSensy Basic), Rs 2,200 (Interakt Growth), Rs 4,899 (Wati Pro) a month on yearly billing, plus Meta's per-message fees (SOURCE: jeedesigns.com, 14 Oct 2026). AiSensy AI agent builder add-on Rs 3,500 a month (SOURCE: theshizz.in, Oct 2026)	Built for marketing and automation. Needs a WhatsApp API number. Bots answer parents with no person in between.
Admission CRMs: ExtraaEdge	Lead management, WhatsApp, IVR, AI lead scoring	Quote on request (SOURCE: Capterra, Oct 2026)	Built mainly for colleges, coaching and universities; setup and price fit larger institutions
School ERPs: Teachmint, Edumaat	Admission module inside a full ERP	Quote on request (SOURCE: G2, SoftwareSuggest)	A full ERP is a big decision for one problem
AdmitAssist	Reads, prioritises with reasons, routes sensitive cases, drafts from school facts, a person always sends	Rs 1,499 per season month, Rs 299 per quiet month	One-hour setup, no new phone number, no automatic replies

Where I could lose: a WhatsApp platform adds "suggested reply with priority" to its inbox. Then AdmitAssist should plug into that platform rather than fight it (V3 in 07). The faculty's point on ecosystems applies: unless I can build my own, align with one.

5. Unit economics tables

Generated by `business/unit_economics.py`. All amounts in rupees, before 18% GST. Token figures are MEASURED from the recorded run of the 48 sample messages (estimated tokenizer: 1 token per 4 Latin characters, 1 per Indic character): read_in_tokens = 1263, read_out_tokens = 251, draft_in_tokens = 801, draft_out_tokens = 114, draft_share = 0.63. Tokens per inquiry include 5% read retries, drafts for the share of messages that get one, 20% automatic and 15% staff rewrites.

1. AI cost per inquiry, by model

Model	Input tokens / inquiry	Output tokens / inquiry	AI cost / inquiry	AI cost for 200 inquiries
gpt-6.1-terra (V1 default)	2007	361	Rs 0.36	Rs 72.39
gpt-6.1-sol (fallback)	2007	361	Rs 0.72	Rs 145
gpt-6-luna (cheapest)	2007	361	Rs 0.04	Rs 7.24
gemini-3.1-flash-lite	2007	361	Rs 0.10	Rs 19.81

2. One school, one month (base case)

Line	Admission-season month (200 inquiries)	Quiet month (30 inquiries)
Price paid by the school	Rs 1,499	Rs 299
AI calls	Rs 72.39	Rs 10.86
Support time	Rs 250	Rs 50.00
Payment collection (2%)	Rs 29.98	Rs 5.98
Storage and bandwidth	Rs 20.00	Rs 20.00
Total cost to serve	Rs 372	Rs 86.84
Contribution	Rs 1,127	Rs 212
Contribution margin	75%	71%

One school per year: revenue Rs 10,788, contribution Rs 8,033 (74%).

3. Winning and setting up one school

Item	Amount
Cost to win one school (4 demos x 1.5 h + travel)	Rs 4,000
Setup and training (3 h + one visit)	Rs 2,000
Payback, in admission-season months	5.3 months

4. How many schools are needed

Goal	Schools needed
Cover the shared platform cost (Rs 2,000/month)	3
Also pay me Rs 50,000/month	78

5. Stress test: admission-season month for one school

Scenario	Revenue	AI cost	Contribution	Margin
Base: 200 inquiries	Rs 1,499	Rs 72.39	Rs 1,127	75%
Heavy school: 600 inquiries	Rs 2,099	Rs 217	Rs 1,320	63%
Very heavy: 1,000 inquiries	Rs 2,899	Rs 362	Rs 1,709	59%
AI price x2 (forced onto fallback model)	Rs 1,499	Rs 145	Rs 1,054	70%
AI price x5 (worst case)	Rs 1,499	Rs 362	Rs 837	56%
Staff ask for 3x more rewrites	Rs 1,499	Rs 83.88	Rs 1,115	74%
Support 1 h per month (B7 target)	Rs 1,499	Rs 72.39	Rs 877	58%
Support 2 h per month	Rs 1,499	Rs 72.39	Rs 377	25%
Support 3 h per month	Rs 1,499	Rs 72.39	Rs -123	-8%
Cheapest model (gpt-6-luna)	Rs 1,499	Rs 7.24	Rs 1,192	80%

Reading: the AI bill is not what can sink this. Support time is. At the 1-hour support target (B7) the margin is still positive, at 3 hours per school per month the school loses money for me, so setup must become self-serve in V2.

6. What the school gets for the money

Item	Amount
Front-office cost per hour (Rs 20,000/month)	Rs 96
Time saved per busy month (6.0 -> 3.6 min x 200)	8.0 hours = Rs 769
Yearly price	Rs 10,788
Extra admissions per year that pay for the tool, budget school (Rs 12,000 fee)	0.90
Same, mid-fee school (Rs 30,000 fee)	0.36
Inquiries per year (base case)	1,380
Conversion lift needed to win 1 extra admission	0.07 percentage points

Reading: time saved alone does not cover the price. The case rests on not losing serious parents. One extra admission a year pays for the tool.

7. Market size (rough)

Measure	Value
Privately managed schools in India	450,000
If every one paid the base yearly price	Rs 485 crore per year
10 schools	revenue Rs 107,880 / contribution Rs 56,328 per year after platform cost
100 schools	revenue Rs 1,078,800 / contribution Rs 779,276 per year after platform cost
500 schools	revenue Rs 5,394,000 / contribution Rs 3,992,379 per year after platform cost

6. Scaling risks and what I do about each

Risk	What happens	Early sign	What I do
Support load grows with schools	Margin is 58% at the 1-hour target and turns negative at 3 hours per school per month (table above)	Support hours above 1 per school per month (B7)	V2 self-serve setup: settings wizard, how-to videos in Telugu and Hindi
Admission season peak for every school at once	All schools busy in the same weeks; AI rate limits	Requests per minute above 50% of the limit shown for the model on my OpenAI account's Limits page	Ask the provider for higher limits by November each year; fallback model; monthly cap per school
Heavy schools	600+ inquiries a month	Overage charges appear	Overage at Rs 2 per inquiry keeps margin at 63% (stress test)
AI price rises 5x	AI cost Rs 362 per busy month	Provider price change	Margin still 56%; switch to the cheapest model that passes the gates
Model quality changes after a provider update	Wrong priorities or missed sensitive messages	Monthly re-run of the evaluation	Pin model versions; FR-23
My father's school is not a fair test	Pilot results higher than other schools will see	Other schools' results worse	Phase 4 schools measured the same way; not counted as revenue
Competition copies the feature	Price pressure	Platforms announce suggested replies	Plug-in route in V3

7. Where the business stands, honestly

At Rs 10,788 a year per school this is a small business unless it reaches hundreds of schools or attaches to a larger platform. 100 schools bring about Rs 10.8 lakh a year in revenue and about Rs 7.8 lakh of contribution after the shared server. That is enough to fund V2 and prove the model. It is not a company yet. The V1 goal is narrower and testable: three schools that owe me nothing pay Rs 1,499 a month by 31 March 2027.